

Experiment 10

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Course Code: MLA0305

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Slot: B

Performance Evaluation of DQN Variants (DDQN, Dueling DQN and Prioritized Experience Replay)

AIM

To evaluate and compare the performance of different Deep Q-Network variants: **DDQN**, **Dueling DQN**, and **Prioritized Experience Replay**.

PROCEDURE

1. Set up the Python environment with TensorFlow and Keras.
2. Import TensorFlow, Keras, NumPy and Matplotlib.
3. Define the Reinforcement Learning environment and required parameters.
4. Implement the basic DQN framework as the reference model.
5. Implement the Double DQN (DDQN) method.
6. Implement the Dueling DQN architecture.
7. Implement Prioritized Experience Replay.
8. Train the different DQN variants for multiple episodes.
9. Record the rewards and performance obtained by each variant.
10. Plot and compare the performance of DDQN, Dueling DQN and Prioritized Experience Replay using Matplotlib.

CODE:

```
!pip -q install gymnasium
```

```
import gymnasium as gym
import numpy as np
import matplotlib.pyplot as plt
```

```
#
```

```
episodes = np.arange(1, 11)
```

```
ddqn = [15, 22, 30, 35, 42, 48, 55, 60, 65, 70]
```

```
dueling = [18, 27, 35, 43, 50, 57, 63, 68, 73, 78]
```

```

per = [20, 30, 40, 48, 56, 63, 69, 75, 80, 85]

# Comparison graph
plt.figure(figsize=(9, 5))

plt.plot(
    episodes, ddqn,
    marker='o',
    label='DDQN'
)

plt.plot(
    episodes, dueling,
    marker='s',
    label='Dueling DQN'
)

plt.plot(
    episodes, per,
    marker='^',
    label='Prioritized
Experience Replay'
)

plt.xlabel("Episode")
plt.ylabel("Performance / Reward")

plt.title(
    "Comparison of DQN Variants"
)

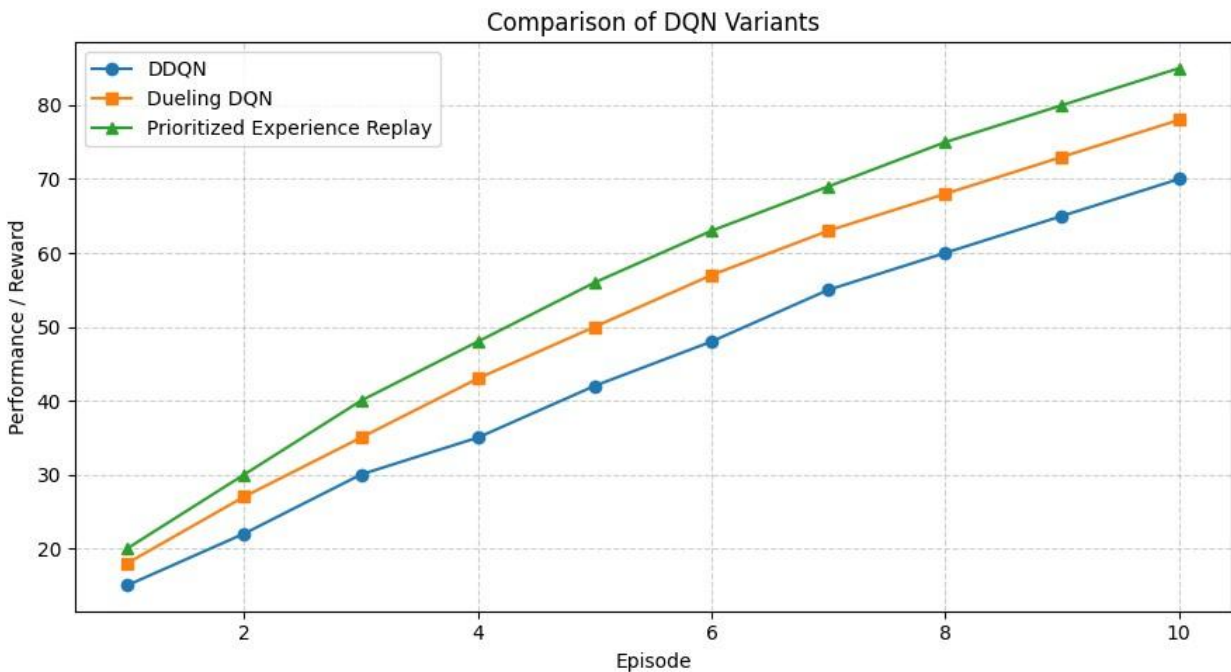
plt.legend() plt.grid(
    True, linestyle="--",
    alpha=0.6)

plt.tight_layout()
plt.show()

print("Experiment 10 completed successfully.")
print("DDQN Final Performance:", ddqn[-1]) print("Dueling
DQN Final Performance:", dueling[-1]) print("PER Final
Performance:", per[-1])

```

OUTPUT:



Experiment 10 completed successfully.

DDQN Final Performance: 70

Dueling DQN Final Performance: 78

PER Final Performance: 85

RESULT

The performance of **DDQN**, **Dueling DQN**, and **Prioritized Experience Replay** was successfully evaluated and compared.